AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 10 · STUDENT LESSON

# If/Then Decisions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

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## Before you start

Week 9: trace every instruction from a stated starting condition.

Plan time to read, investigate and explain. Keep predictions separate from observations.

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STUDENT LESSON · 1 OF 18

## Your mission

> Start with the target: Evaluate a condition using the current state and trace only the selected branch. Try the recall before opening notes.

### Your learning target

I can evaluate a **condition** using the current state and trace only the selected **branch**.

I can distinguish inclusive and strict boundaries using values below, at and above a **threshold**.

I can design a **boundary test** and handle an unknown **input** without silently treating it as false.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

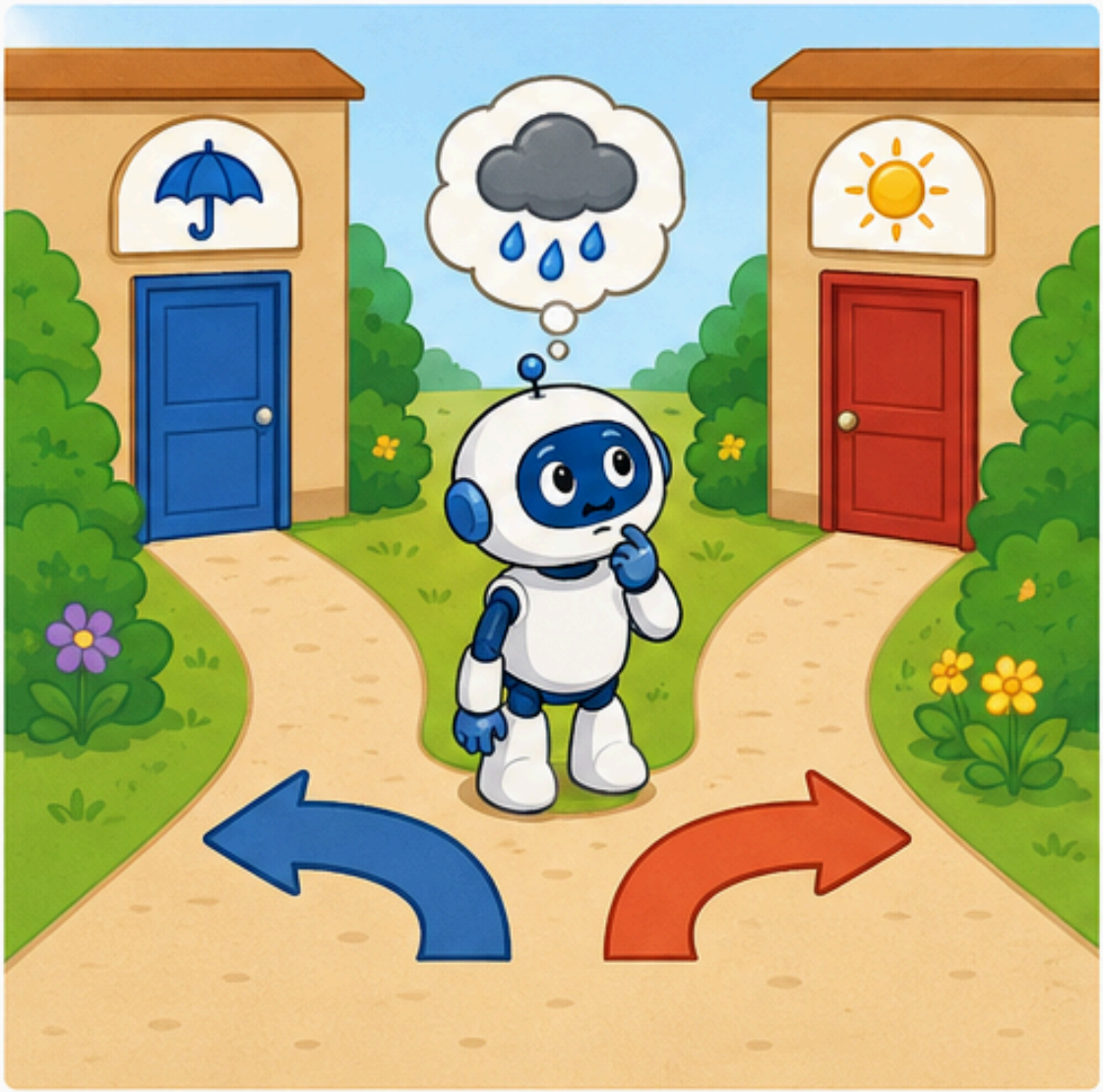
Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

**Condition**s make an algorithm **branch** when information changes.



### Your target

LEARNING TARGET A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

Learner name:

**Readiness for the connected explanation**

Score starts at 4. Add 2 to score, then double it. Which value is doubled, and what is the final score?

**This week's end goal**

By the end, I can trace exactly one **branch** and test a boundary that includes equality.

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STUDENT LESSON · 2 OF 18

## Notice the evidence

> Investigate the supplied If/Then Decisions case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

**Look Closely Before You Explain**

Pictures give clues; **evidence** tells us what the clues mean.



### Start with this puzzle

MYSTERY A clothing helper follows different paths depending on temperature, rain, and wind **condition**s.

**What do you notice?**

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

### Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

**A closer explanation**

A paper game updates a score with this procedure: add 2 to score; if score is at least 6, subtract 3 from score; otherwise add 4 to score; report score and stop. The if and otherwise instructions form one if/**else** decision.

Exactly one of those **branch**es runs each time this decision is reached.

A **condition** asks whether a statement is true for the current state. At least 6 means 6 or any larger number. If score is 6, the **condition** is true. If score is 5, it is false.

The **condition** itself does not change score; the chosen **branch** performs the update.

Start score at 4. First add 2, giving 6. Now test the current 6: the **condition** is true, so subtract 3 and report 3. Testing the original 4 would choose the wrong **branch**.

Write the state immediately before the **condition** so the **evidence** for the choice is visible.

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STUDENT LESSON · 3 OF 18

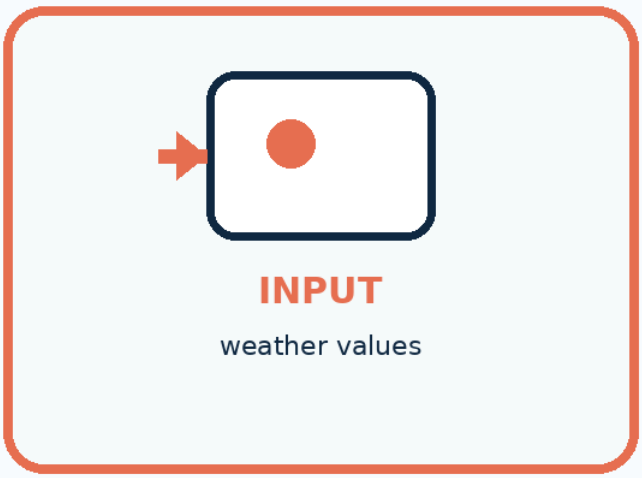
## See how it works

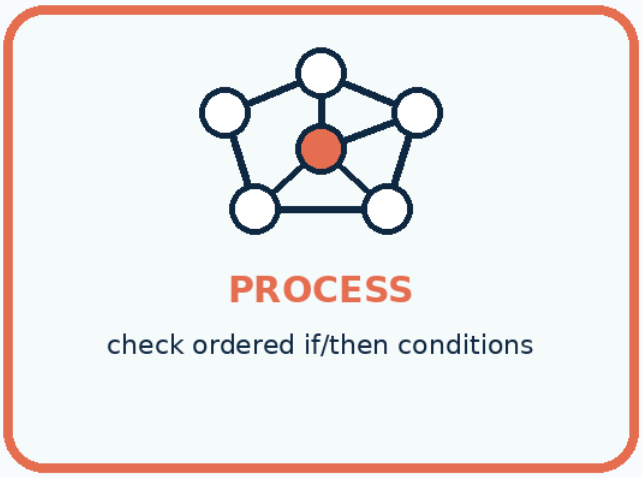
> Follow the mechanism for If/Then Decisions. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

**Input → Process → Output**

Follow the information instead of imagining magic inside the machine.









**INPUT** weather values

**PROCESS** check ordered if/then **condition**s

**OUTPUT** clothing suggestion and reason

### Key idea

CORE IDEA **Condition**s make an algorithm **branch** when information changes.

### How the mechanism works

UNDER THE HOOD A **condition**al evaluates a Boolean expression and selects a **branch**. Order matters when **condition**s overlap. Complete decision logic needs explicit handling for boundary values and an **else** path.

Draw a fresh example with all three stages:

### A closer explanation

Compare the three values immediately around the **threshold**. These are the values when the **condition** is reached, after the initial add-2 instruction.

Testing only a value far above 6 would not show whether equality was handled correctly.

### Three checks around the threshold

| SCORE AT THE CONDITION | CHOOSE ONE BRANCH | SCORE AFTER THE BRANCH |
| --- | --- | --- |
| Score is 5 | At least 6 is false so add 4 | Score becomes 9 |
| Score is 6 | At least 6 is true so subtract 3 | Score becomes 3 |
| Score is 7 | At least 6 is true so subtract 3 | Score becomes 4 |

At least includes equality. The middle row is essential for checking this boundary. Read each row as a separate run; do not pass the result of one row into the next.

Above 6 is different: it means strictly greater than 6, so 6 itself does not pass. At most 6 means 6 or smaller. Below 6 means strictly smaller, so it excludes 6. Say the equality case aloud before using one of these phrases.

A **condition** can be true even when its chosen action makes the number smaller. True describes the test result; it does not mean good, bigger, or a successful game. The written instructions define the action.

### Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

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STUDENT LESSON · 4 OF 18

## Compare the cases

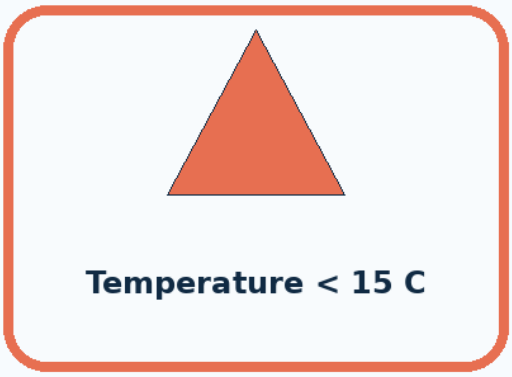
> Compare the supplied cases for If/Then Decisions. Identify the detail that changes the conclusion.

PICTURE GALLERY

### Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.












### The case to investigate

CASE 1 — Rain = yes Clue: then choose raincoat. Meaning: clear binary **branch**.

### The case to investigate

CASE 2 — Temperature < 15 C Clue: then choose warm layer. Meaning: numeric **threshold**.

### The case to investigate

CASE 3 — Exactly 15 C Clue: needs boundary rule. Meaning: edge case.

### The case to investigate

CASE 4 — Rain and cold Clue: two **condition**s overlap. Meaning: order or combined rule needed.

### Keep this boundary clear

BOUNDARY Never hide an unhandled case between two **threshold**s.

Which two pictures are easiest to confuse? What **evidence** separates them?

**Change one thing in the pictured case**

### Change one thing

Change: Replace at least 6 with above 6.

### Predicted result

**Prediction**: Only the run starting at 4 changes: it reports 12.

### Why the result follows

Reason: Updated value 6 fails a strict above comparison and is doubled.

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STUDENT LESSON · 5 OF 18

## Words that unlock the idea

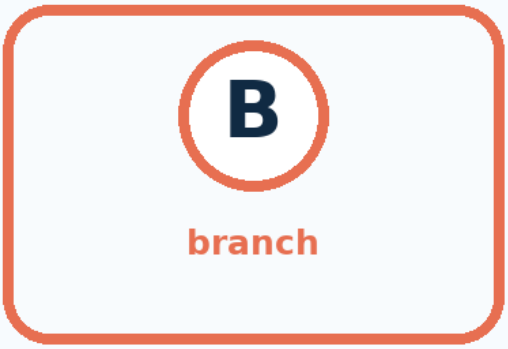
> Use the vocabulary of If/Then Decisions to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

### Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.












| Word | Meaning | My example |
| --- | --- | --- |
| condition | question evaluated as true or false | |
| branch | path selected by a condition | |

| Word | Meaning | My example |
|---|---|---|
| threshold | value where a decision changes | |
| else | path used when earlier conditions are false | |

### Misconception detective

> **Find the unsupported claim**
>
> CLAIM TO REPAIR If one example works, the **branch** is complete.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

### Words used in the closer explanation

| Word | Meaning |
|---|---|
| Inclusive boundary | A boundary whose equality case is included |
| Boundary test | A test using a value at or close to a point where behaviour can change |

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STUDENT LESSON · 6 OF 18

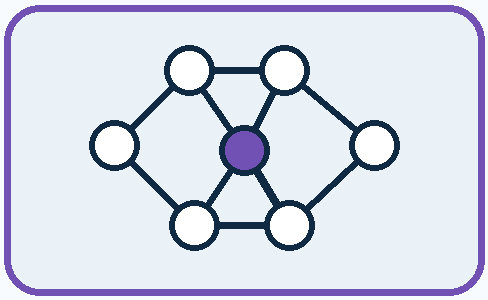
# Follow a worked example

> Follow the worked trace for If/Then Decisions, predict a step, then compare with its explanation.

WORKED EXAMPLE

### Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



### The case to investigate

CASE A clothing helper follows different paths depending on temperature, rain, and wind **condition**s.

### Step 1

STEP 1 — OBSERVE weather values

### Step 2

STEP 2 — TRACE check ordered if/then **condition**s

### Step 3

STEP 3 — CONCLUDE clothing suggestion and reason

### Model answer

MODEL REASONING Each test follows the stated **condition** order. The exactly-at-**threshold** case reveals whether the rule uses less than or less-than-or-equal. Rain-plus-cold needs combined **output**s or a clear priority.

### A claim to avoid

DO NOT OVERCLAIM Never hide an unhandled case between two **threshold**s.

What new **evidence** could change this answer?

#### A closer explanation

| Initial score | State and condition after adding 2 | Branch and final score |
|---|---|---|
| 3 | 5 is not at least 6 | Add 4 and report 9 |
| 4 | 6 is at least 6 | Subtract 3 and report 3 |
| 5 | 7 is at least 6 | Subtract 3 and report 4 |

### Worked explanation · If/Then Decisions

In the first run, the false **branch** increases 5 to 9. That does not make the earlier decision run again. The procedure has already chosen and carried out the otherwise **branch**, so it moves on to report and stop.

Subtracting 3 afterward would incorrectly run both **branch**es and report 6.

This rule applies to one if/**else** decision. A later, separate **condition** would be a new test of the state at that later point. Do not invent an extra test or loop because the value crossed the **threshold**.

Follow only the instructions in the procedure.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: If/Then Decisions

A fictional counter begins with a number, adds 2, then chooses one **branch**.

### The facts we will use

#### Supplied facts

If the updated value is at least 6, subtract 1; otherwise double it.

#### Supplied facts

Report once and stop. Use separate starting values 3,4,5.

AI ACADEMY / AI-1/10

# Inclusive branching trace

| | Start | After +2 | At least 6? | Report |
|---|---|---|---|---|
| Run 1 | 3 | 5 | false | 10 |
| Run 2 | 4 | 6 | true | 5 |
| Run 3 | 5 | 7 | true | 6 |

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/10

# If/Then Decisions

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Update first

Starts 3,4,5 become 5,6,7.



## 2. Test current value

5>=6 is false; 6>=6 and 7>=6 are true.



## 3. Choose one branch

Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6.



## 4. Report

The three independent runs report 10,5,6.

**CHANGE ONE THING**

Replace at least 6 with above 6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Update first | Starts 3,4,5 become 5,6,7. |
| 2. Test current value | 5≥6 is false; 6≥6 and 7≥6 are true. |
| 3. Choose one branch | Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6. |
| 4. Report | The three independent runs report 10,5,6. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### Why the result follows

The **condition** is evaluated after adding 2. Equality belongs to the at-least **branch**, so starting at 4 checks the boundary.

Once a **branch** finishes, the program reports; it does not reconsider the **condition** just because doubling 5 produced 10.

### What this example does not establish

#### Keep the stated limit

A large reported value need not mean the true **branch** ran; **branch** choice depends on the earlier value.

#### Keep the stated limit

Read the labelled picture: Which **input** detects the difference between the two written **condition**s? Explain its entire run.

#### Keep the stated limit

Change one thing: Replace at least 6 with above 6. Predict the result before checking the explanation. What remains the same?

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STUDENT LESSON · 7 OF 18

## Find and repair a mistake

> Find the unsupported step in the If/Then Decisions claim and explain a repair using the supplied evidence.

COMPARE

### Same Result, Different Reason

A comparison helps reveal the true concept boundary.

| Case | Evidence | What it shows |
| --- | --- | --- |
| Rain = yes | then choose raincoat | clear binary branch |
| Temperature < 15 C | then choose warm layer | numeric threshold |
| Exactly 15 C | needs boundary rule | edge case |
| Rain and cold | two conditions overlap | order or combined rule needed |

### Repair two tempting claims

#### Challenge claim A

CLAIM A If one example works, the **branch** is complete.

Repair:

#### Challenge claim B

CLAIM B Changing **branch** order cannot change the result.

Repair:

#### A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

**A closer explanation**

### Check the claim against the facts

Our examples so far use known whole-number scores. If score is unreadable when the test is reached, we cannot establish whether it is at least 6. Unknown does not mean false, and a blank is not automatically zero.

For this paper exercise, pause the procedure and request the missing value before selecting a **branch**.

### Check the claim against the facts

A complete procedure must specify what to do when required **input** is unavailable. Another system might define a separate fallback, but it must be written explicitly.

Testing below, at and above the boundary checks comparison behaviour; it does not replace a missing-**input** policy or prove every possible implementation correct.

**A tester checks only scores 2 and 10 at the condition and says the at-least-6 comparison is correct. Which additional value is most useful for detecting an accidental above-6 comparison, and why**

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STUDENT LESSON · 8 OF 18

## Practise together

Practise If/Then Decisions with support, then explain the deciding step in your own words.

GUIDED PRACTICE

### Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: **Condition**s make an algorithm **branch** when information changes.

Use a fresh example to explain **condition**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Changing **branch** order cannot change the result.

Invent one edge case and name the safe response.

#### Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

### A closer explanation

Suppose a copied procedure says above 6 instead of at least 6. A score of 5 at the **condition** takes the false **branch** under both versions. A score of 7 takes the true **branch** under both.

But exactly 6 separates them: the intended version subtracts 3 and gives 3; the copied version adds 4 and gives 10.

> **Choose an initial score that reaches exactly 6 after the first add-2 instruction. Trace both the intended at-least version and the copied above version. Explain why this test reveals an error that the neighbouring values can miss**

Record the initial value, the value at the **condition**, the true or false result, and the chosen update. A final number alone may not reveal whether the test or the arithmetic was wrong.

Keep the expected behaviour tied to the stated policy instead of changing it to match a result.

### Practise with less help: Read four boundary phrases

Use workbook W2 for the connected example "**Condition**s and boundary cases". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
| --- | --- | --- |
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for **Condition**s and boundary cases. Proposed explanation: "The **condition** reads the starting value". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Which **input** detects the difference between the two written **condition**s? Explain its entire run.

**My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

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STUDENT LESSON · 9 OF 18

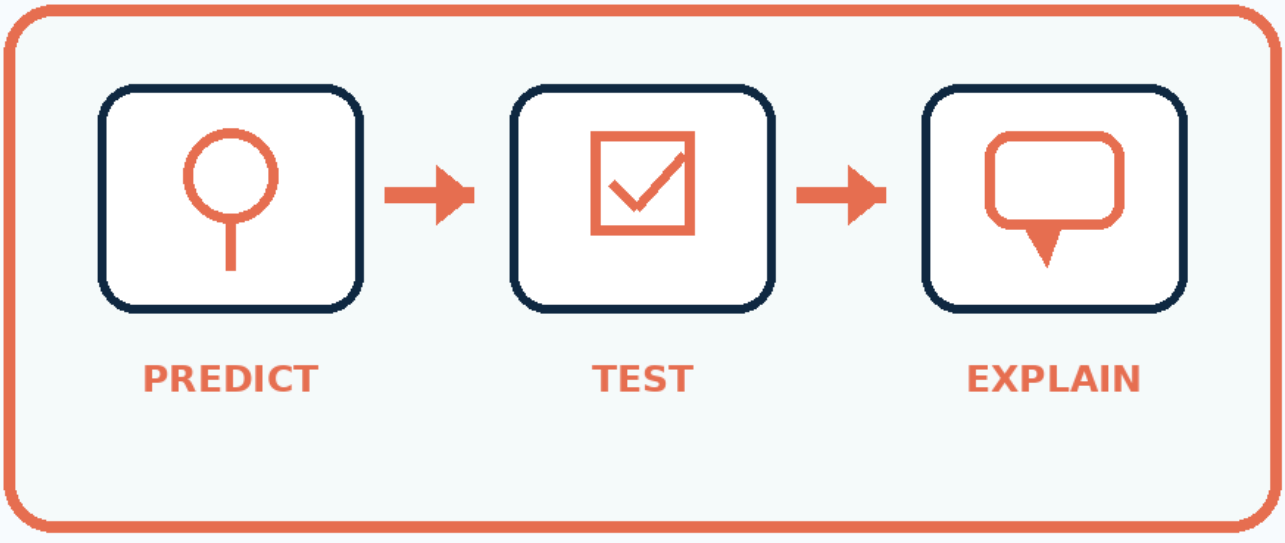
## Run the investigation

> Investigate If/Then Decisions. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

### Weather Wardrobe Branches

Predict first, test honestly, and explain what the **evidence** means.



### 1. Predict

State your **prediction** before the result is known.

### 2. Investigate

Complete the task: Build a weather-clothing rule and test four cases.

### 3. Compare cases

Keep a normal case and an edge case clearly separated.

### 4. Record honestly

Record what happened—even when it does not match your **prediction**.

### 5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

| Case | Prediction | Observed result | What it means |
|---|---|---|---|
| Normal | | | |
| Edge | | | |

### Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

### Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

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STUDENT LESSON · 10 OF 18

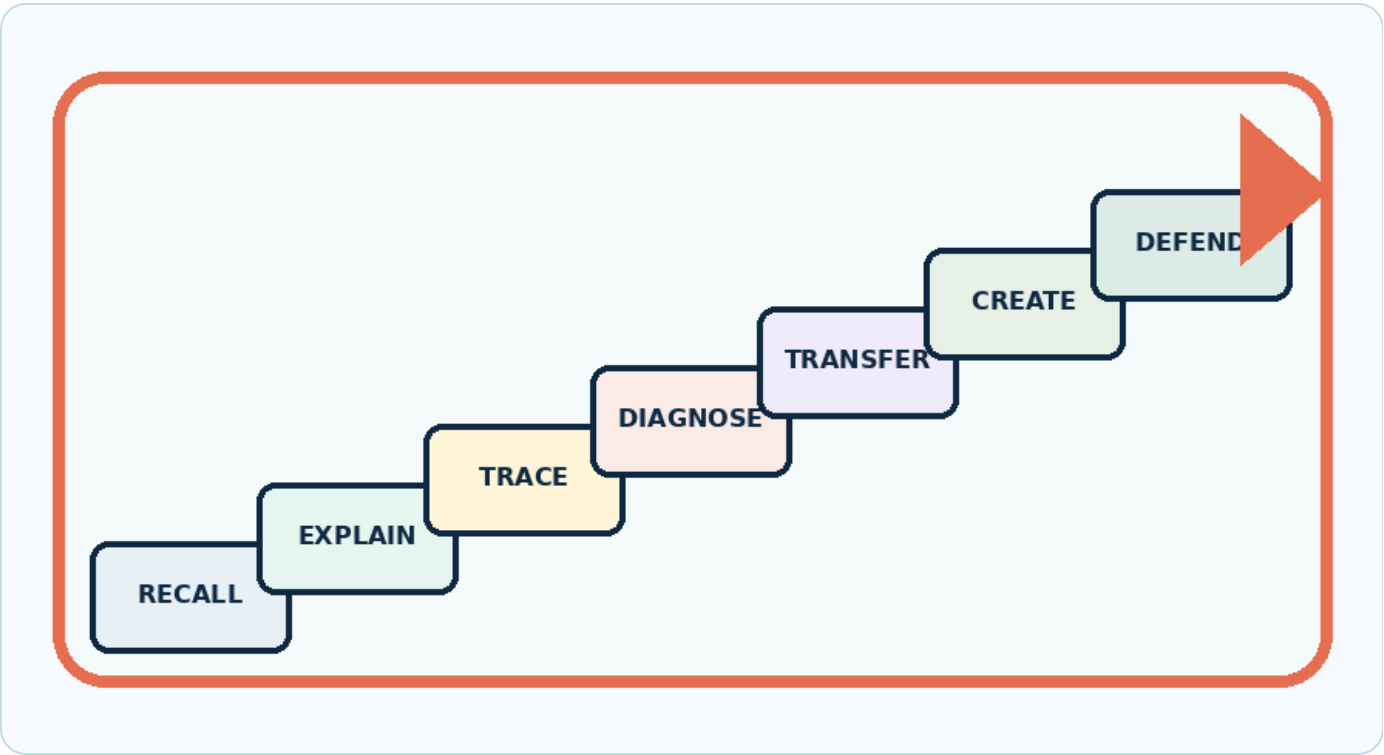
## Build your understanding

> Move from recall to tracing, diagnosis and transfer in If/Then Decisions. Explain why each step is justified.

PRACTICE LADDER

### Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RECALL Define **condition**.

EXPLAIN **Condition**s make an algorithm **branch** when information changes.

TRACE Show the **input**, **process**, and **output**.

DIAGNOSE Repair: If one example works, the **branch** is complete.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

### When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

STUDENT LESSON · 11 OF 18

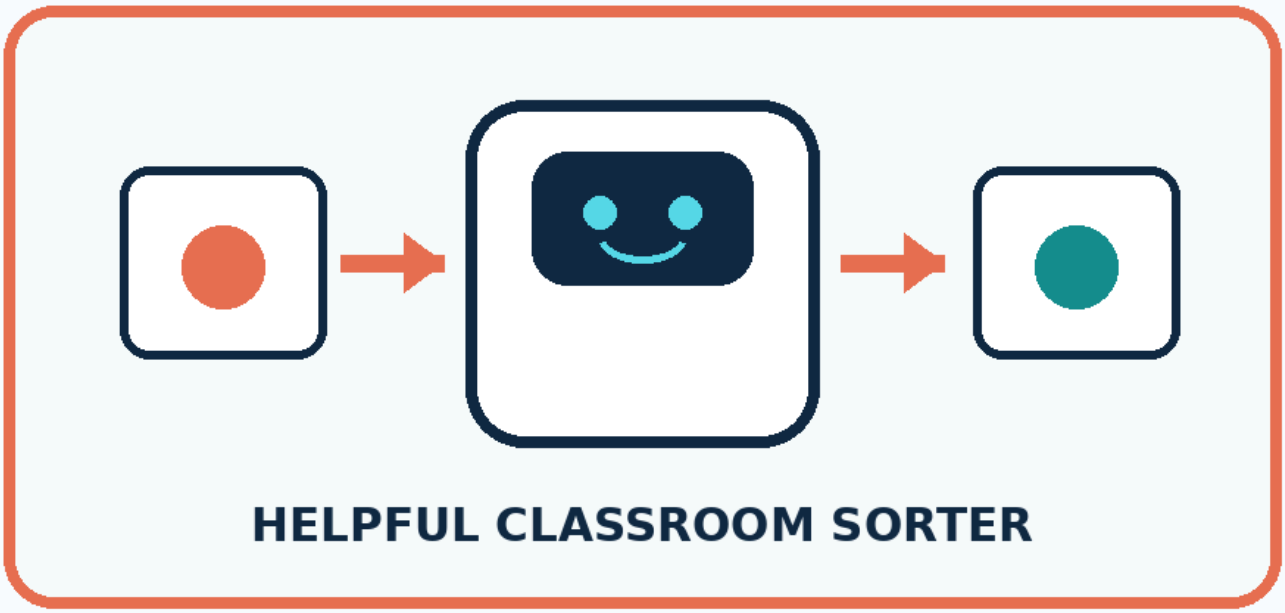
# Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

### Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.




THIS WEEK'S CHECKPOINT Add one transparent if/then **branch** and one **else** path to the Classroom Sorter.

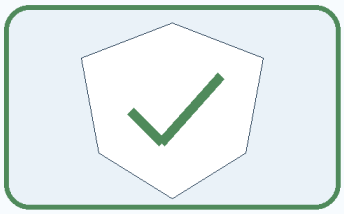
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



**Responsible design**

RESPONSIBLE DESIGN Never hide an unhandled case between two **threshold**s.

My project decision:

### Evidence for your project

Include equality examples when reviewing sorter **condition**s, and trace from original **input** through **output**.

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STUDENT LESSON · 12 OF 18

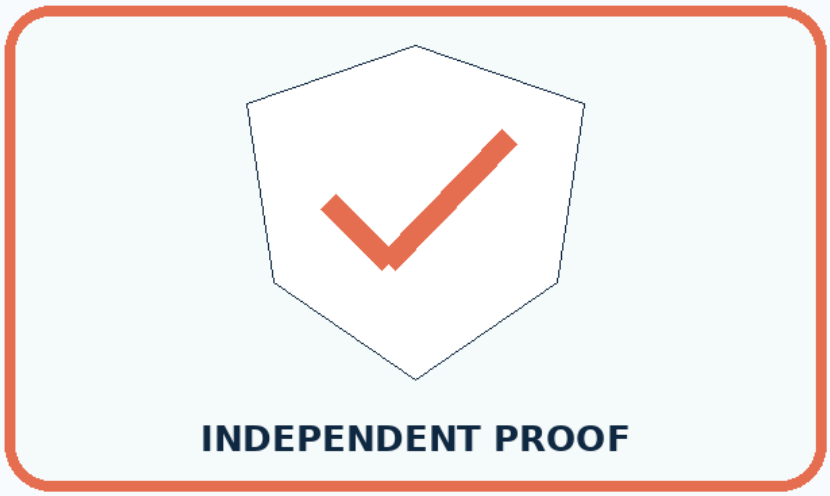
## Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

### Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.




### What success looks like

REQUIRED PROOF Build a clothing rule and trace sunny, rainy, cold, and exactly-at-

**threshold** cases.

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

### Check your explanation

SELF-CHECK Each test follows the stated **condition** order. The exactly-at-**threshold** case reveals whether the rule uses less than or less-than-or-equal. Rain-plus-cold needs combined **output**s or a clear priority.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

### Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Repeating Efficiently.

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STUDENT LESSON · 13 OF 18

## Study a complete case

Revisit the complete If/Then Decisions case for supported practice. Repeated examples do not become fresh assessments.

## Week 10 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

### The situation

If a card has a star, send it to A; otherwise send it to B.

### The question

Sort a star, a circle, and a card with both shapes.

### Worked explanation

Star and mixed cards go to A because both have a star. The circle goes to B. The test is has a star, not has only a star.

### Explain it in your own words

> Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

### Trace the supplied case visually

### Does this card have a star?

| Evidence or stage | Value or result |
| --- | --- |
| Star only | Yes → tray A |
| Circle only | No → tray B |

| Evidence or stage | Value or result |
| --- | --- |
| Star and circle | Yes → tray A |

If a card has a star, send it to A; otherwise send it to B.

Sort a star, a circle, and a card with both shapes.

Star and mixed cards go to A because both have a star. The circle goes to B. The test is has a star, not has only a star.

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STUDENT LESSON · 14 OF 18

## Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Transfer the method to a new display

A paper brightness display adds 3 to its starting number. If the current brightness is at most 7, subtract 2; otherwise add 1. Report and stop. Trace separate runs from starting values 3, 4 and 5.

Identify the run that tests equality. Then show how its result changes if at most 7 is accidentally written as below 7.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 10 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Use this exact rule: if a card has more than three dots, place it in tray A; otherwise use tray B. Trace cards with 2, 3 and 4 dots.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

### Connect to your project

Add one transparent if/then **branch** and one **else** path to the Classroom Sorter.

The **evidence** I will keep

**Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

### Transfer practice from the original programme

A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

**Explain the mechanism, evidence and one limit.**

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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STUDENT LESSON · 15 OF 18

## Your lesson route

Use this map to navigate If/Then Decisions, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional counter begins with a number, adds 2, then chooses one **branch**.

| By the end, I can | How I will show it |
|---|---|
| Evaluate a condition using the current state and trace only the selected branch | Explain the deciding step, show a trace or test, and state one limit. |
| Distinguish inclusive and strict boundaries using values below, at and above a threshold | Explain the deciding step, show a trace or test, and state one limit. |
| Design a boundary test and handle an unknown input without silently treating it as false | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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STUDENT LESSON · 16 OF 18

## Finish and return

> Explain If/Then Decisions without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Evaluate a **condition** using the current state and trace only the selected **branch**

**My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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STUDENT LESSON · 17 OF 18

## Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional counter begins with a number, adds 2, then chooses one **branch**.

**Practice 1: explain the deciding evidence**

Which labelled step matches this detail? The three independent runs report 10,5,6.

1. Update first

2. Report

3. Choose one **branch**

4. Test current value

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Report The three independent runs report 10,5,6. The **condition** is evaluated after adding 2.

Equality belongs to the at-least **branch**, so starting at 4 checks the boundary. Once a **branch** finishes, the program reports; it does not reconsider the **condition** just because doubling 5 produced 10.

## Practice 2: explain the deciding evidence

### Changed case · predict first

Changed-case practice: Replace at least 6 with above 6. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Only the run starting at 4 changes: it reports 12.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed **condition** and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Only the run starting at 4 changes: it reports 12. Updated value 6 fails a strict above comparison and is doubled.

**Repair a missing building block**

Review Algorithms as Recipes (ai-1/9). Goal: Trace an ordered algorithm using the current value after every instruction. Short practice: Use the arrows to explain where the two zero-start runs first differ..

Return to this week and explain what you changed.

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version..

Return to this week and explain what you changed.

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used labels and which stage used only the new measurement..

Return to this week and explain what you changed.

**Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

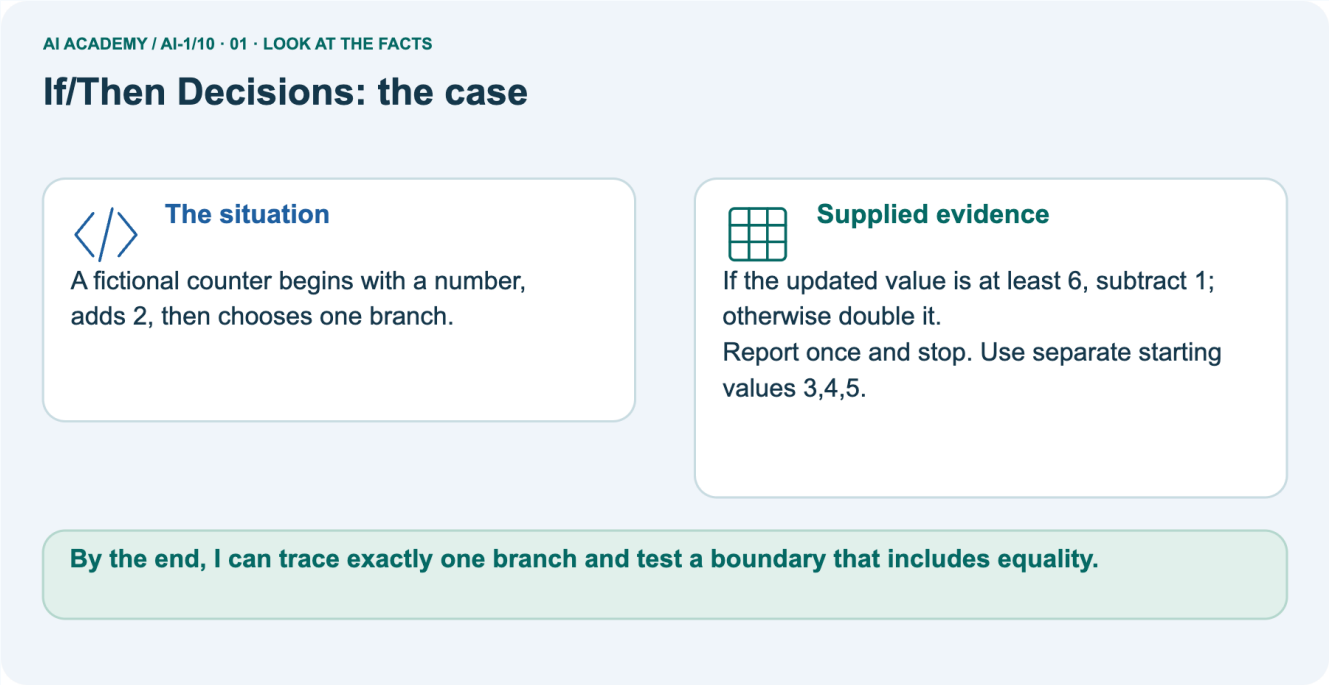
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**STUDENT LESSON · 18 OF 18**

## Visual concept journey

> Explain each If/Then Decisions graphic using its labels, deciding evidence and relevant limitation.

## 01. If/Then Decisions: the case

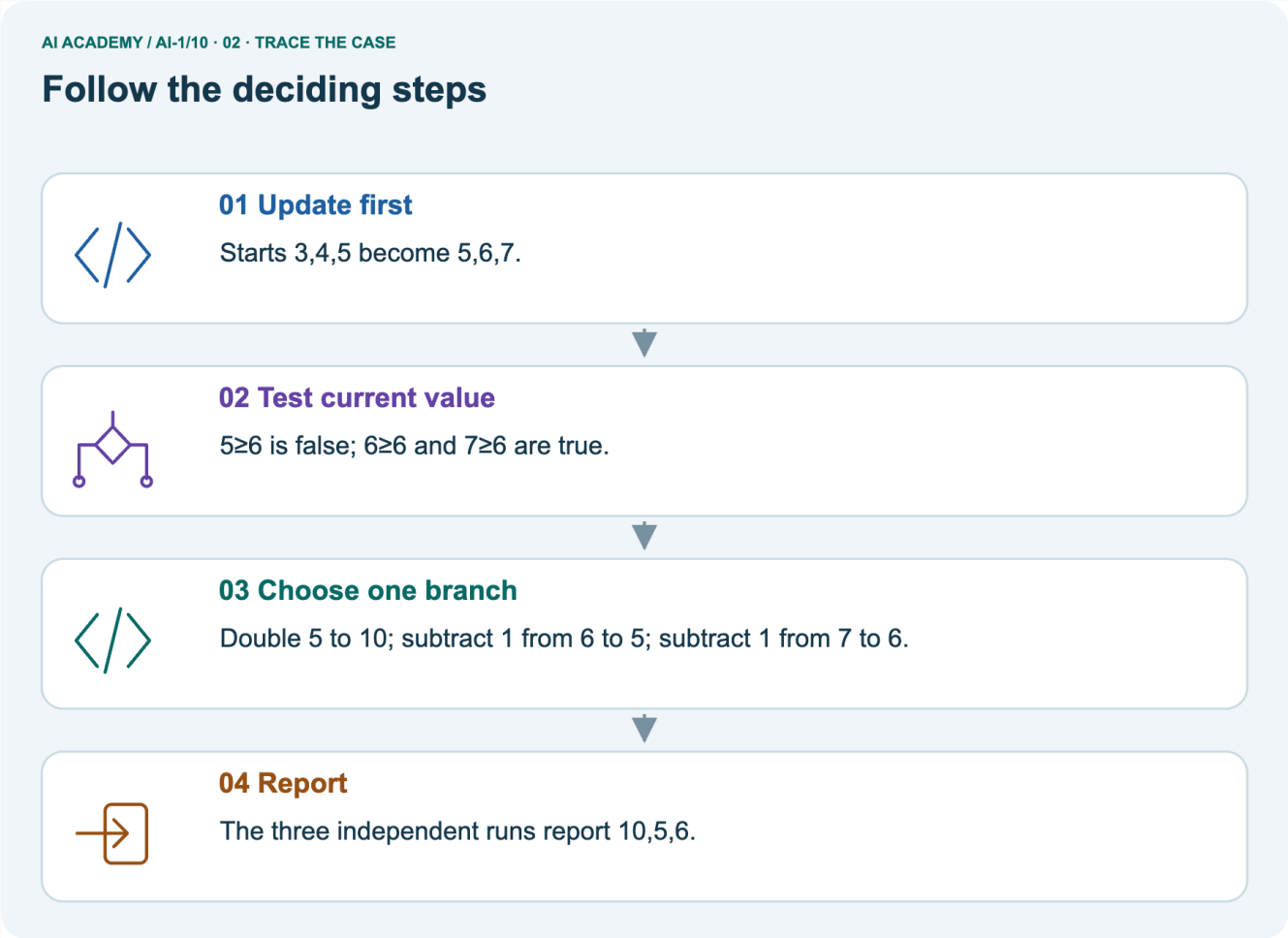



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional counter begins with a number, adds 2, then chooses one **branch**. Supplied facts: If the updated value is at least 6, subtract 1; otherwise double it. Report once and stop. Use separate starting values 3,4,5.

> Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps

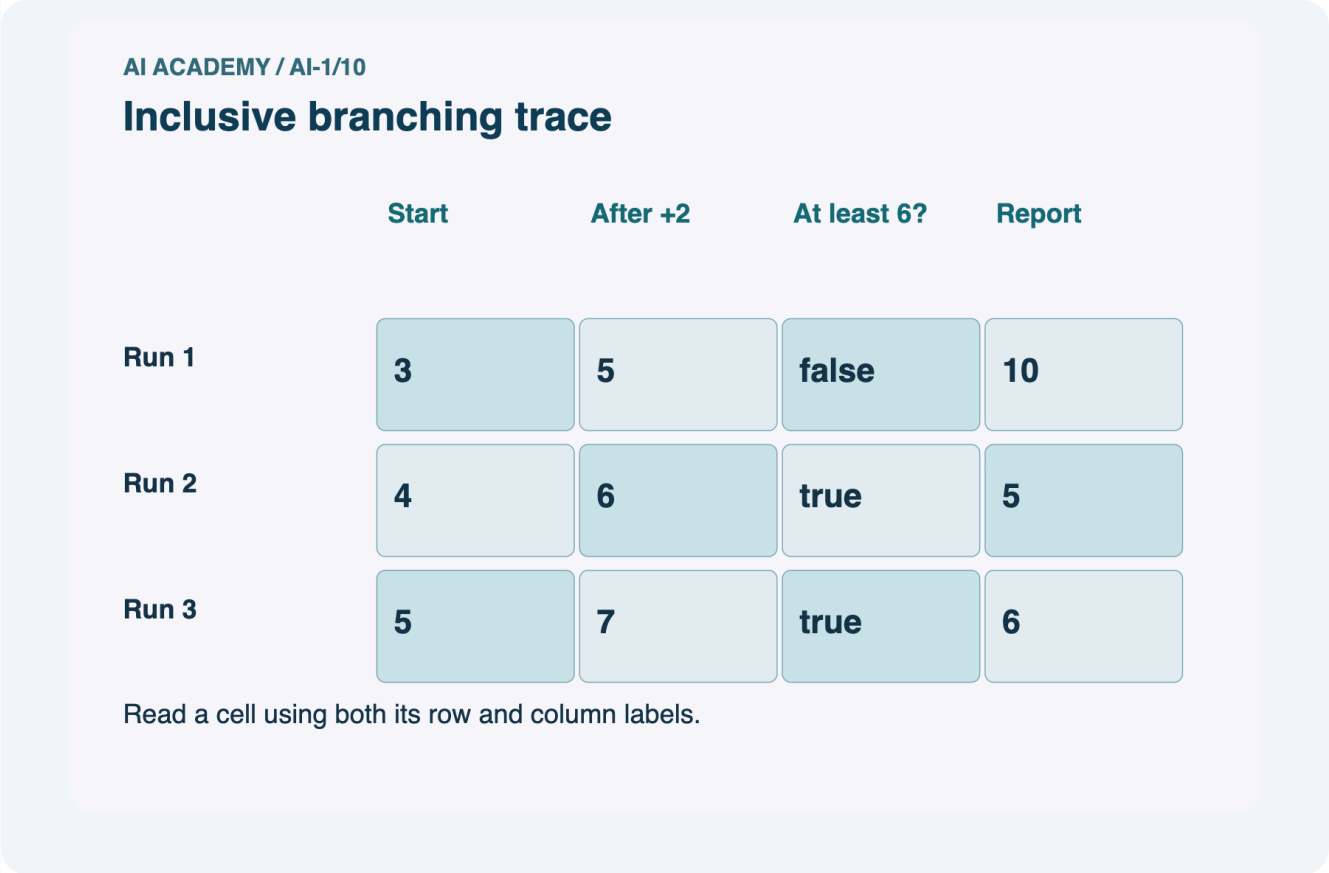



Follow the labelled picture with the matching explanation. Open the vector diagram

Update first: Starts 3,4,5 become 5,6,7. → Test current value: 5≥6 is false; 6≥6 and 7≥6 are true. → Choose one **branch**: Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6. → Report: The three independent runs report 10,5,6.

> Point to each numbered stage and explain which recorded **input** it uses.

### 03. Inclusive branching trace

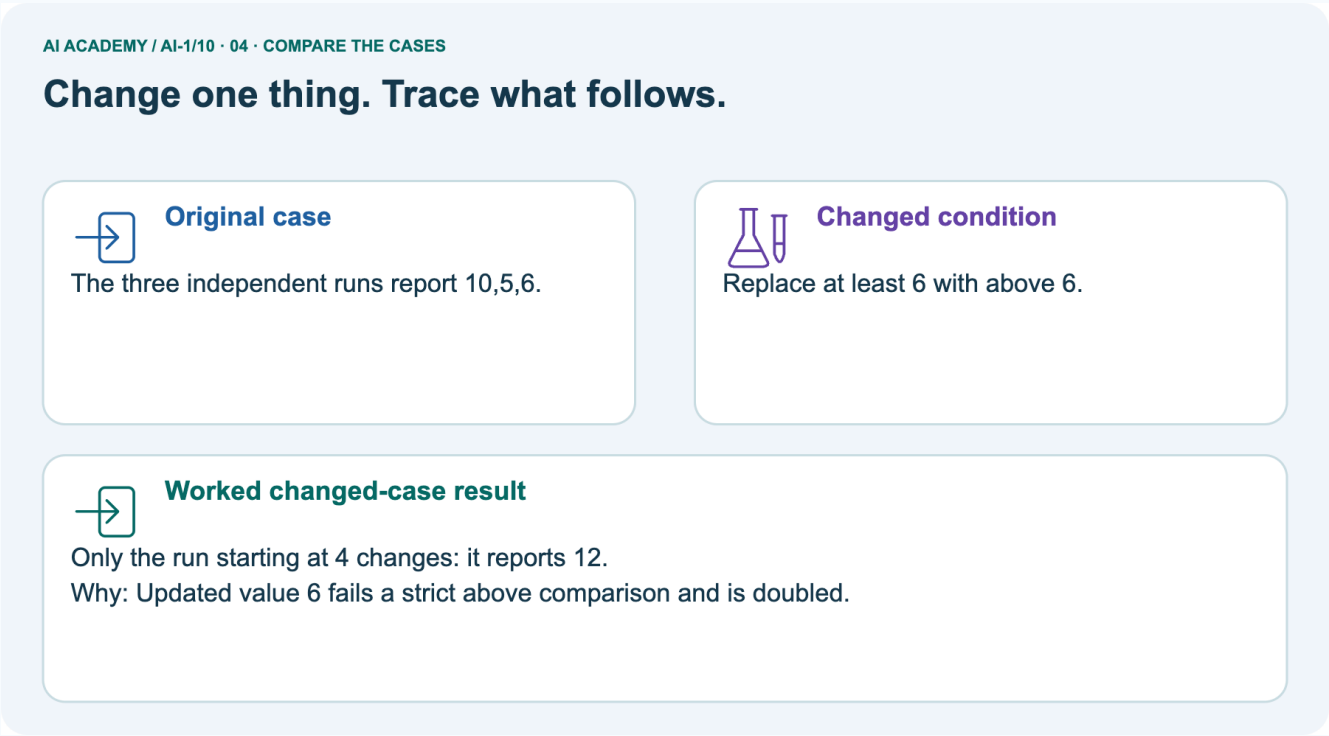



Follow the labelled picture with the matching explanation. Open the vector diagram

Inclusive **branch**ing trace. Run 1: Start = 3, After +2 = 5, At least 6? = false, Report = 10. Run 2: Start = 4, After +2 = 6, At least 6? = true, Report = 5. Run 3: Start = 5, After +2 = 7, At least 6? = true, Report = 6.

> Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



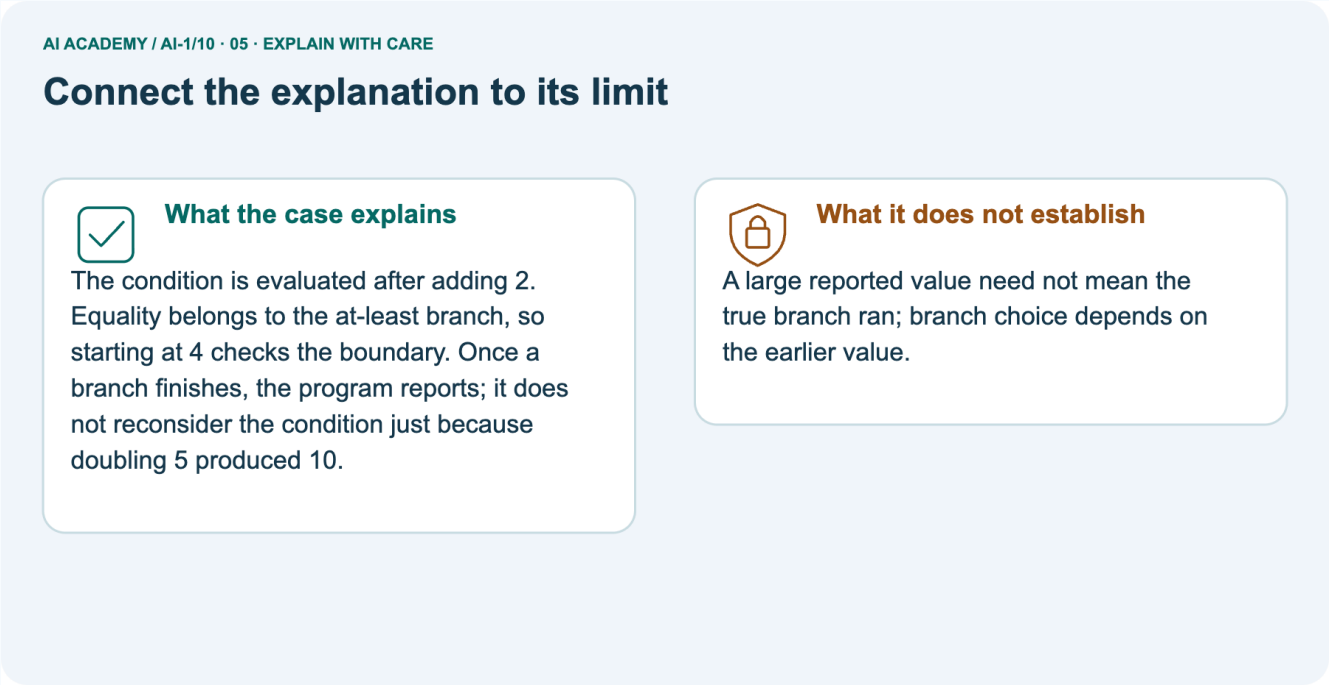

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The three independent runs report 10,5,6. Changed **condition**: Replace at least 6 with above 6. Expected result: Only the run starting at 4 changes: it reports 12.

Reason: Updated value 6 fails a strict above comparison and is doubled.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

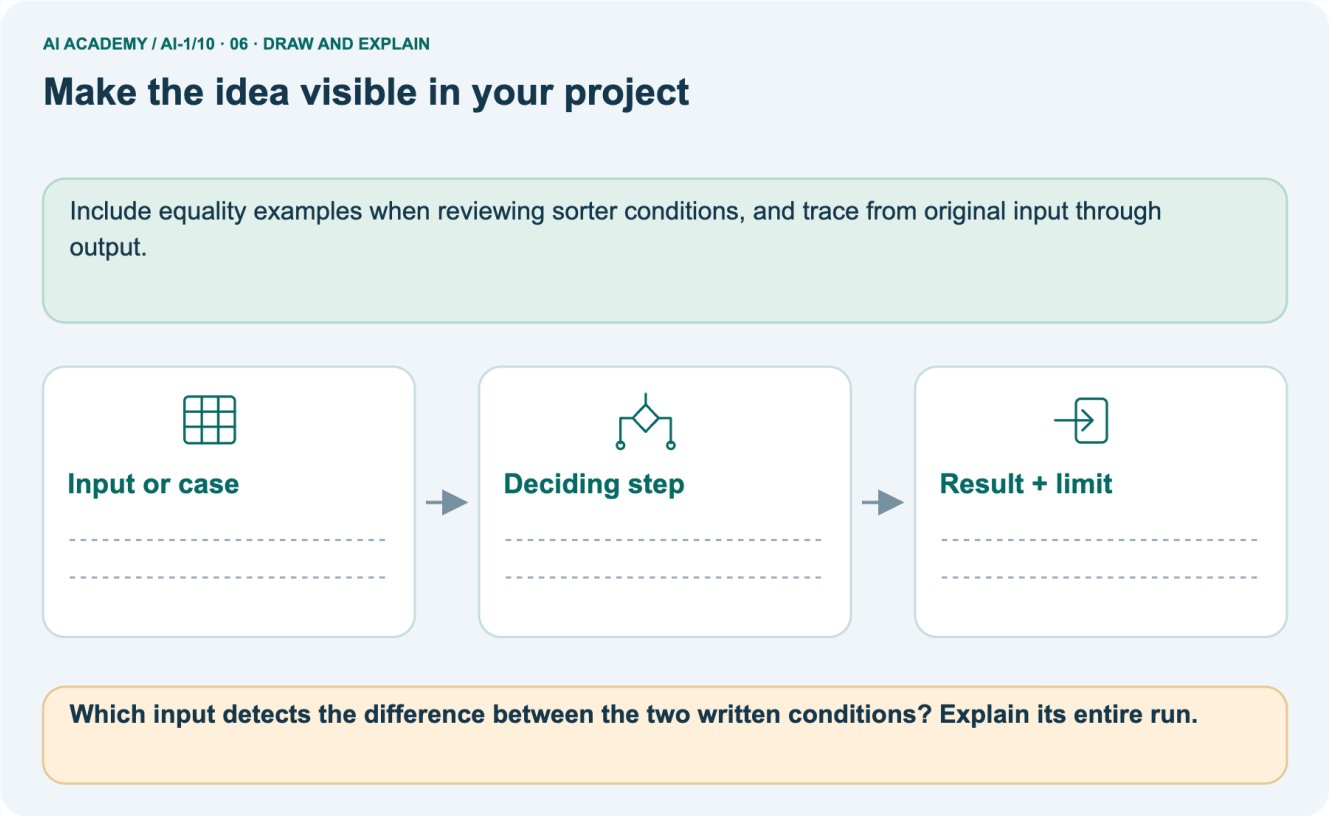
The **condition** is evaluated after adding 2. Equality belongs to the at-least **branch**, so starting at 4 checks the boundary.

Once a **branch** finishes, the program reports; it does not reconsider the **condition** just because doubling 5 produced 10.

Limit: A large reported value need not mean the true **branch** ran; **branch** choice depends on the earlier value.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Include equality examples when reviewing sorter **condition**s, and trace from original **input** through **output**. Draw your **input**, deciding step, result and limit.

Supported question: Which **input** detects the difference between the two written **condition**s? Explain its entire run.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.